# DASE7506 Advanced Machine Learning Methods

Mini Project 2 Report

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**Abstract**

This project studies four-candidate answer selection on GSM8K using the provided candidate texts and embeddings. Candidates with the same parsed final numerical answer are treated as answer groups. The method combines embedding-based candidate scoring and shallow format and numerical answer features through group-level calibration, alongside a model trained directly for answer-group selection. The final system averages the group probabilities from two calibrated predictors and one directly optimized model with fixed weights of 0.25, 0.25, and 0.50. Out-of-fold experiments show improvements from group-aware decision rules and numerical answer features, while additional question and reasoning-related signals do not consistently improve the stronger system. The frozen model achieves 387 correct selections out of 467 questions (82.87% accuracy) on the official test set.

## 1 Introduction

Large language models can generate multiple candidate solutions to mathematical word problems, but selecting a correct answer remains a separate challenge. In this project, each GSM8K question is accompanied by four generated solutions and their embeddings. Since multiple candidates may share the same final numerical answer, the task involves selecting among answer groups rather than treating candidate positions as mutually exclusive classes. Simple majority voting captures answer agreement but does not fully exploit the available candidate information.

The proposed system combines embedding-based candidate scoring with group-level decision making. Two embedding models trained under different objectives are paired with a shared shallow feature model and separately calibrated at the answer-group level. A third model directly optimizes group selection. Their resulting probabilities are combined through fixed-weight fusion.

The experiments evaluate these components using out-of-fold comparisons and controlled ablations. Additional signals from question text, intermediate calculations, and candidate representations are also examined to determine whether they improve an established group-aware baseline. The analysis focuses on both predictive performance and the reliability of incremental improvements.

## 2 Problem Setup

### 2.1 Task Definition

Each GSM8K question is accompanied by four candidate solutions generated by Qwen3-0.6B and a 1024-dimensional embedding for each candidate. The task is to select one of the four candidates by returning its original index $\hat{i} \in \{0, 1, 2, 3\}$. A prediction is considered correct if

the selected candidate's final numerical answer matches the reference answer under the official parser and scorer. The evaluation focuses on the final answer rather than the correctness of intermediate reasoning steps.

The official parser converts equivalent numerical representations, such as decimals and fractions, into canonical values. Let $a_i$ denote the parsed final answer of candidate $i$. Candidates sharing the same valid answer are grouped together:

$$I_g = \{i \mid a_i = g\}, \qquad n_g = |I_g|, \tag{1}$$

where $I_g$ is the set of candidate indices belonging to answer group $g$, and $n_g$ is its size. Each question contains at most four valid answer groups, and candidates with unparseable final answers are excluded from group-level selection.

Since multiple candidates may share the correct numerical answer, their correctness labels are not necessarily mutually exclusive. A group-based approach therefore selects a final numerical answer first and then maps the selected group back to an original candidate index.

### 2.2 Evaluation protocol

The training set contains 4,755 questions, including 2,114 disagreement cases with at least two distinct valid numerical answers. An early internal split divides the training set into 3,804 fitting questions and 951 development questions for initial experiments. Subsequent out-of-fold evaluations use the full training set. Table 1 summarizes the dataset sizes.

Table 1: Dataset sizes and disagreement counts. Fit and Dev are subsets of Training.

| Dataset | Total | Disagreement |
|---|---|---|
| Training (full) | 4,755 | 2,114 |
| Fit subset | 3,804 | 1,691 |
| Dev subset | 951 | 423 |
| Validation | 798 | 355 |
| Test | 467 | 467 |

Selection accuracy measures the proportion of questions for which the selected candidate matches the reference numerical answer. Disagreement accuracy is the primary development metric because it focuses on competing answer groups.

Model comparisons primarily use five-fold out-of-fold (OOF) evaluation. Within each outer training fold, three-fold inner OOF predictions are generated to fit score normalizers and group calibrators. The outer held-out questions are excluded from fitting these components. Three partitions of the same training set are examined rather than independent datasets.

Paired accuracy differences are summarized using descriptive 95% question-level bootstrap intervals. Disagreement negative log-likelihood (NLL) measures the mean negative natural-log probability assigned to the correct answer group. Repeated model selection may still introduce bias. The official test set was evaluated only once using the frozen final system.

# 3 Method

## 3.1 Candidate Scoring

Three candidate-level scoring models are used in the final system: B, C, and F. The first two learn from the provided embeddings under different training objectives, while F uses shallow features extracted from candidate texts and their final numerical answers.

**Model B (BCE-dis).** Model B applies a linear classifier to each standardized 1024-dimensional candidate embedding. Its output $b_i$ represents a candidate-level logit trained with binary cross-entropy (BCE). Each valid candidate is labeled according to whether its parsed final answer matches the reference answer, allowing multiple positive candidates within the same question. Training is restricted to disagreement questions, where distinct numerical answers compete for selection.

**Model C (Group-CE).** Model C also uses a linear classifier on standardized candidate embeddings, but optimizes group-level cross-entropy over the full training set. For each question, its candidate logits $c_i$ are averaged within each valid answer group:

$$\bar{c}_g = \frac{1}{n_g} \sum_{i \in I_g} c_i. \tag{2}$$

A softmax over these group scores is used to optimize the probability of the correct answer group. Unlike B, this objective directly considers competition between distinct numerical answers during training.

**Model F (Format/Final).** Model F uses 20 handcrafted features: 12 describing candidate format and 8 describing the parsed final answer. Format features include text length, line structure, answer markers, and punctuation statistics. Final-answer features describe numerical properties such as sign, magnitude, integer status, and fraction components. These standardized features are processed by a shallow neural network with one 32-unit ReLU hidden layer, trained using BCE on valid candidates from the full training set. The features capture answer structure and numerical patterns without explicitly verifying mathematical reasoning.

The resulting logits $b_i$, $c_i$, and $f_i$ are used by the subsequent group-level decision models. B and C provide different learning objectives, while F supplies shared auxiliary information to both calibrated branches.

## 3.2 Group Calibration

Models B and C produce candidate-level scores, which are combined separately with the shared model F to form two calibrated predictors, BF and CF. Each predictor converts its two scoring streams into answer-group probabilities through a group-level calibration model.

For each scoring stream, candidate logits are centered by the mean score of valid candidates within the question and scaled by a standard deviation estimated from inner out-of-fold (OOF) predictions. The normalized scores $z_i$ are aggregated within each answer group using Log-Mean-Exp (LME):

$$\mathrm{LME}_g(z) = \log \left( \frac{1}{n_g} \sum_{i \in I_g} \exp(z_i) \right). \tag{3}$$

The temperature is fixed at 1. Unlike simple averaging, LME gives greater influence to higher-scoring candidates while normalizing by group size.

For either BF or CF, let $u_g$ and $v_g$ denote the LME scores from its two scoring streams. The calibration model also considers group frequency $f_g = n_g/4$, the normalized number of valid groups $k = K/4$, and two score margins $m_u$ and $m_v$. Each margin is calculated from the difference between the two highest group means, rather than LME scores, and clipped to $[0, 4]$.

These quantities form an 11-dimensional feature vector:

$$\begin{aligned}\tilde{\boldsymbol{\phi}}_g = [&u_g, v_g, u_g m_u, v_g m_u, u_g m_v, v_g m_v,\\ &u_g f_g, v_g f_g, f_g, u_g k, v_g k]^\top.\end{aligned} \tag{4}$$

The features are divided by fixed feature-wise RMS scales to obtain $\boldsymbol{\phi}_g$. The SD and RMS statistics are fitted using inner OOF disagreement predictions. RMS retains the valid-candidate weighting of the original mean-based feature map and is not refitted using LME features. A linear calibrator then assigns a probability to each valid answer group:

$$P(g) = \frac{\exp(\boldsymbol{\theta}^\top \boldsymbol{\phi}_g)}{\sum_{h \in \mathcal{G}} \exp(\boldsymbol{\theta}^\top \boldsymbol{\phi}_h)}, \tag{5}$$

where $\mathcal{G}$ is the set of valid answer groups. The calibration parameters are optimized using group-level cross-entropy with L2 regularization.

BF and CF use separate calibration parameters, fitted on inner OOF disagreement predictions. This allows their different candidate-scoring objectives to produce distinct group-level probability distributions while sharing the same auxiliary model F.

### 3.3 Direct Optimization

Direct-LME optimizes answer-group selection without a separate calibration stage. Each valid candidate is represented by a 1044-dimensional vector $\mathbf{x}_i$, combining its standardized embedding and 20 Format/Final features. A linear model produces the candidate logit $u_i = \mathbf{w}^\top \mathbf{x}_i$, which is aggregated into the group score

$$L_g = \log\left(\frac{1}{n_g}\sum_{i \in I_g} \exp(u_i)\right) + \alpha \log n_g, \tag{6}$$

where $\alpha$ is a learnable coefficient controlling the influence of group frequency. The group probability is obtained as $P_D(g) = \mathrm{softmax}(L)_g$ over valid groups.

Training minimizes the correct group's cross-entropy on disagreement questions, with L2 regularization on $\mathbf{w}$ and a penalty encouraging $\alpha$ toward 1. Optimization follows a fixed two-stage procedure: a group-mean variant is trained first, and its parameters initialize the LME model.

Unlike BF and CF, Direct-LME learns group-level predictions within a single optimization objective. Its probabilities form the third input to the final fusion.

### 3.4 Probability Fusion

The final system combines three answer-group probability distributions: $P_{BF}$ and $P_{CF}$ from the separately calibrated predictors, and $P_D$ from Direct-LME. All three distributions are defined

over the same valid answer groups. The final probability is calculated as

$$P_{\text{final}}(g) = 0.25P_{BF}(g) + 0.25P_{CF}(g) + 0.50P_D(g). \tag{7}$$

The fusion operates on complete group probabilities rather than raw candidate logits. The weights are fixed across questions, without an additional gating or calibration model. The answer group with the highest fused probability is selected, and its earliest original candidate index is returned. If no valid final answer can be parsed, the prediction defaults to index 0.

For final training, three-fold OOF base-model predictions from the full training set are used to fit separate BF and CF calibrators on disagreement questions. The base models are then refitted using their respective training populations: disagreement questions for B and all training questions for C and F. Preprocessing statistics are fitted on the full permitted training partition. Direct-LME is trained separately on disagreement questions. All components are frozen before official test evaluation.

Figure 1 illustrates the final architecture.

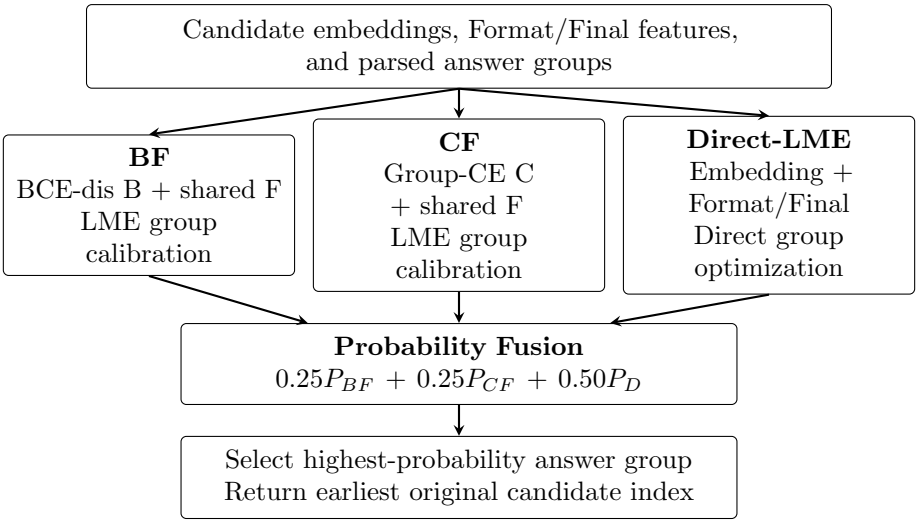



Figure 1: Final prediction architecture with a shared Format/Final model.

# 4 Experiments

## 4.1 Baselines

Initial baselines were evaluated on the internal development split containing 423 disagreement questions. Table 2 compares answer-frequency voting, embedding-based classifiers, and more expressive scoring models.

Embedding-based scoring substantially outperformed majority voting. B2-masked used the same trained classifier as B2 but excluded candidates whose final numerical answers could not be parsed during selection. This improved performance without retraining. Compared with B1, B2-masked corrected 82 questions while introducing 38 new errors, yielding a net gain of 44.

More expressive models did not consistently perform better. R1 fell below B2-masked, while N1 achieved the highest development accuracy but showed a substantial training–development performance gap. These results demonstrate the value of embedding-based scoring over simple answer voting, but provide limited evidence that increasing model complexity alone leads to

Table 2: Baseline results on Internal Dev disagreement questions.

| Method | Correct / 423 |
|---|---|
| B0: Fixed index 0 | 213 |
| B1: Majority voting | 284 |
| B2: Embedding logistic regression | 325 |
| B2-masked | 328 |
| R1: Pairwise ranking | 310 |
| N1: Shallow neural scorer | 333 |

more reliable selection. Subsequent experiments therefore focus on training objectives and answer-group decision mechanisms.

### 4.2 Group-Level Ablations

Table 3 summarizes the main ablation results on the original five-fold OOF evaluation, covering 2,114 disagreement questions. The experiments examine training objectives, training populations, group calibration, and shallow candidate features.

Table 3: Group-level ablations on Train OOF disagreement questions.

| Configuration | Correct / 2114 |
|---|---|
| BCE-all (group-mean decoding) | 1602 |
| Group-CE-all (group-mean decoding) | 1611 |
| BCE-dis (group-mean decoding) | 1641 |
| BCE-dis + mean calibration | 1691 |
| Format-only, mean calibrated | 1695 |
| Final-only, mean calibrated | 1722 |
| Format/Final, mean calibrated | 1722 |
| Format/Final, LME calibrated (BF) | 1730 |

The three mean-calibrated feature configurations are alternative extensions of the 1691-correct baseline, while the final row replaces mean aggregation with LME in the full Format/Final system. Restricting BCE training to disagreement questions improved the linear embedding model from 1602 to 1641 correct selections, with 133 corrected and 94 newly incorrect predictions. Group calibration further increased the result to 1691 by incorporating answer-group frequency and score margins. These comparisons suggest that both training population and group-level decision rules affect selection performance.

Shallow feature ablations show that final-answer numerical properties provide most of the observed benefit. Adding Final-only features increased the calibrated result from 1691 to 1722, whereas Format-only features reached 1695. Combining both feature sets also produced 1722 correct selections, indicating no additional net gain from Format features under this configuration.

Replacing mean-based group aggregation with LME increased the full BF system from 1722 to 1730 correct selections. This change corrected 22 predictions while introducing 14 new errors. The improvement was modest, with a descriptive paired confidence interval crossing zero. Overall, the results support group-aware calibration and numerical answer features as useful components, while the additional benefit of LME remains less certain.

### 4.3 Fusion Analysis

The three prediction branches were compared using three five-fold OOF partitions of the training set. Table 4 reports the number of correct selections among the same 2,114 disagreement questions under each partition.

Table 4: Fusion results across Train OOF partitions (correct / 2114).

| Model | Original | Repartition | Third |
|---|---|---|---|
| BF | 1730 | 1718 | 1721 |
| CF | 1727 | 1718 | 1719 |
| BF + CF (Phase13) | 1735 | 1729 | 1725 |
| Direct-MEAN | 1720 | 1719 | 1718 |
| Direct-LME | 1727 | 1723 | 1713 |
| Phase13 + Direct-MEAN | 1735 | 1738 | 1730 |
| Final fusion (Phase14) | 1736 | 1733 | 1737 |

Although BF and CF share the same auxiliary feature model, they differ in training objectives, training populations, and calibration paths. On the original OOF partition, BF correctly selected 75 questions missed by CF, while CF correctly selected 72 questions missed by BF. Averaging their complete group probabilities improved upon BF alone by 5, 11, and 4 correct selections across the three partitions, indicating modest predictive complementarity.

Adding Direct-LME further increased the correct selections from 1735 to 1736, 1729 to 1733, and 1725 to 1737. The corresponding paired comparisons showed 45/44, 44/40, and 61/49 corrected/newly incorrect predictions. The gains were small relative to the number of changed predictions, and the descriptive paired confidence intervals included zero. Direct-LME was not consistently stronger as a standalone model, but contributed complementary predictions when fused with the calibrated branches.

Probability quality was also examined using disagreement negative log-likelihood (NLL). The final fusion reduced NLL relative to Phase13 from 0.4305 to 0.4232, 0.4323 to 0.4250, and 0.4343 to 0.4298 across the three partitions. These results support retaining the fixed fusion as a modest improvement in both selection accuracy and probability quality, without establishing a statistically significant accuracy gain. Since all three partitions contain the same questions, they are treated as repeated split checks rather than independent evaluations.

### 4.4 Additional Signals

Additional information from questions, intermediate calculations, and candidate texts was examined to identify potential improvements beyond embedding-based scoring. Table 5 summarizes representative controlled comparisons on the original five-fold OOF partition.

Question-number features showed clear independent predictive value, reaching 1191 correct selections compared with 977 for Final-only features, 1007 for mismatched inputs, and 1065 for the capacity control. However, incorporating these features into the stronger 1730-correct system produced no additional correct selections, while the capacity control reached 1734.

The remaining experiments showed more limited benefits. Contextual embeddings reached 1500 correct selections, compared with 1485 for mismatched inputs and 1476 for the same-capacity baseline, but the improvement was not robust. Intermediate arithmetic relations reached 1522, slightly below the final-answer structural control at 1525.

Table 5: Additional-signal comparisons (correct / 2114).

| Signal | Reference control | Mismatched | Capacity | True signal |
|---|---|---|---|---|
| Question numbers (standalone) | Final-only: 977 | 1007 | 1065 | 1191 |
| Question numbers (strong model) | Baseline: 1730 | 1729 | 1734 | 1730 |
| Intermediate arithmetic | Final structure: 1525 | 1511 | – | 1522 |
| Embedding context | Same-capacity: 1476 | 1485 | – | 1500 |

A separate numerical-role text model also demonstrated predictive value. Its standalone performance exceeded a generic text control across three OOF partitions. In an experimental residual extension, numerical-role features were used to correct the log probabilities of the frozen fusion. This reduced NLL, while changes in correct selections were +6, +1, and -6 across the three partitions. The extension was not adopted in the final system. These results highlight the difference between useful predictive information and consistent improvements in selection accuracy.

### 4.5 Official Results

Table 6 reports the official validation and test results of the frozen final system.

Table 6: Official evaluation results of the final system.

| Evaluation | Correct / Total | Accuracy (%) |
|---|---|---|
| Validation (Disagreement) | 303 / 355 | 85.35 |
| Validation (Overall) | 746 / 798 | 93.48 |
| Test | 387 / 467 | 82.87 |

On validation disagreement questions, the final system achieved 303/355 correct selections, compared with 300/355 for BF. The paired comparison included 10 corrected and seven newly incorrect predictions, resulting in a net gain of three. The validation set had also been used during development, and the two complete systems used different inner-calibration splits. The descriptive paired 95% interval for the accuracy difference was $[-1.41, 3.10]$ percentage points.

## 5 Discussion

The experiments highlight the importance of answer-group modeling in candidate selection. Training on disagreement questions, group calibration, and final-answer numerical features contributed to improved selection accuracy. The final fusion also benefited from differences between the calibrated and directly optimized predictors. Although the accuracy gains were modest across the three OOF partitions, the consistent reductions in negative log-likelihood suggest improvements in the combined probability estimates.

Another finding concerns the use of additional information. Question-number features showed independent predictive value, while numerical-role text representations improved probability quality. However, incorporating these signals into the stronger system produced mixed results in selection accuracy. This suggests that extracting useful information and integrating it effectively

into an existing decision system are separate challenges. More reliable methods for identifying when additional signals should influence the final prediction remain a promising direction.

The evaluation also has limitations. The three OOF partitions reuse the same training questions, and iterative experimentation may introduce model-selection bias. The reported comparisons therefore reflect the observed development performance, while the frozen official test evaluation provides the final assessment of the selected system.

# 6 Reproducibility

The final system was trained using the provided training data without external datasets or additional pretrained models. Models B, C, and F used Adam for 900 updates, with a batch size of 128 questions, a learning rate of 0.001, and L2 regularization of 0.0001 (seed 20261029). Three-fold OOF predictions were generated from the full training set to fit separate BF and CF calibrators on disagreement questions, using L2 regularization of 0.01 (final calibration seed 20261130). The base models were then refitted using their respective training populations, while Direct-LME was trained separately on disagreement questions with L2 regularization of 0.01. Group calibrators and Direct-LME used L-BFGS-B with a maximum of 300 iterations per stage. The frozen model version, `phase14-full-train-v1`, includes the weights, preprocessing statistics, calibration parameters, and configuration files required for inference.

The uncompressed inference assets total 61,839 bytes (0.059 MiB), below the 16 MiB limit. On a local macOS ARM64 system, the complete official test prediction process took 0.564 seconds, followed by 0.076 seconds for independent scoring, totaling 0.640 seconds. The processes were executed sequentially with the thread pool limited to one thread. These measurements reflect the local environment rather than the reference Xeon platform.

Generative AI tools, including ChatGPT and Codex, assisted with research planning, code implementation, debugging, experiment analysis, and report preparation. All reported experimental results are based on saved evaluation records and the official scoring procedure. No external language model inference was used during model training or prediction.

# 7 Conclusion

This project developed a lightweight answer-selection system for four-candidate GSM8K problems by combining candidate-level scoring with answer-group decision making. Group calibration and numerical answer features provided substantial improvements in out-of-fold experiments, while the fusion of BF, CF, and Direct-LME offered smaller, consistent gains across the examined partitions. The frozen system correctly selected 387 out of 467 questions on the official test set. Further experiments identified useful signals in question numbers and candidate texts, highlighting the potential for more effective integration of additional information into group-aware prediction systems.